Stochastic Deep Supervision for Gallery-Consistent Early-Exit Face Recognition

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github.com/gurayozgur/SD-ViT

One face recognition model, many compute budgets

- Face recognition runs from **edge processors to datacenter accelerators**, under budgets that differ by orders of magnitude.
- A Vision Transformer processes every input through **all L blocks**, at the same cost, whatever the device.
- Compact architectures, compression and distillation each produce **another static model**: covering several budgets means training, storing and switching whole networks.

Goal: one trained network that can stop at **any block depth**, chosen by the compute budget.

Early exit meets open-set recognition

Closed-set early exit (BranchyNet, MSDNet, SDN)

- every exit has its own classifier over a fixed label set
- exits need not share a representation

Face recognition

- open-set: a probe is matched to a **gallery of enrolled embeddings** by cosine similarity
- per-exit heads give **incompatible spaces**

Gallery consistency: every depth must emit embeddings in **one shared space**, compatible with a gallery **enrolled once at full depth**.

Galleries can be massive, distributed or privacy-constrained: re-enrolling them per depth is not an option.

A standard ViT has no usable early exits

Baseline ViT-S/12 on WebFace4M, intermediate blocks read through the final head.

IJB-C TAR@FAR= 10^{-4}	exit 3	exit 6	exit 12
probe and gallery at exit k	11.05	42.22	96.89
probe at exit k, gallery at full depth	**0.02**	**34.63**	96.89

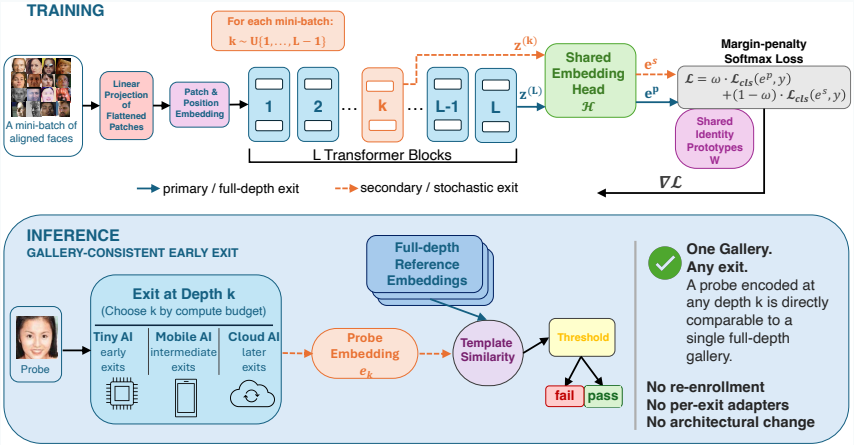
- Only the final embedding is supervised; intermediate blocks get gradients through backpropagation alone and are **never required to be embeddings**.
- This read-out is exactly training-free ViT-FREE.

The alternatives give up the single gallery

- **Per-exit heads** (closed-set multi-exit ViTs, ViT-FREE$_{FT}$): each exit has its own space, so the gallery must be **re-enrolled at every depth**.
- **Adaptation layers** between depths (FitNets-style): break the single-embedding-space assumption enrollment pipelines rely on.
- **Dedicated networks per budget**: N budgets need N trainings and N galleries that are not interchangeable.

We want: **one network, one gallery, a usable exit at every block depth**, with no teacher and no extra layers at inference.

SD-ViT: stochastic shared-classifier deep supervision



At every iteration, besides the full-depth output, read out **one uniformly sampled block**, decode it with the **same head** $\mathcal{H}$, and score it against the **same prototypes** W under the same margin loss.

The objective

$$e^{\mathrm{p}} = \mathcal{H}(z^{(L)}), \qquad e^{\mathrm{s}} = \mathcal{H}(z^{(k)}), \qquad k \sim \mathcal{U}\{1, \ldots, L-1\}$$

$$\mathcal{L} = \omega\, \mathcal{L}_{\mathrm{cls}}(e^{\mathrm{p}}, y) + (1-\omega)\, \mathcal{L}_{\mathrm{cls}}(e^{\mathrm{s}}, y), \qquad \omega = 0.7$$

- $\mathcal{H}$: shared LayerNorm + projection ϕ; a ViT keeps $N \times d$ tokens at every block, so **one head fits every depth**.
- $\mathcal{L}_{\mathrm{cls}}$: any margin-penalty softmax loss (AdaFace; ArcFace behaves the same).
- Both terms are supervised by the identity label: **no teacher, no detached target, no divergence term**. Deep supervision through a shared classifier, an alignment rather than a transfer.
- Sampling one block per step is a **single-sample estimator** of supervising all exits.

Why do the exits become comparable?

1. The shared head

primary

- one LN, one projection ϕ, one set of BN statistics must make **final and sampled tokens** discriminative
- $z^{(L)} = z^{(k)} + \sum_{l>k} \Delta_l$: two snapshots of **one residual stream**
- through the same ϕ, later blocks are pushed to **refine** what ϕ reads at depth k, not re-encode it

2. The shared prototypes

secondary

- the margin loss drives $\hat{e}$ toward $\hat{w}_y$
- one W pulls identity y at every depth toward **one center**

Neither exit teaches the other; **they learn together**. The controls later attribute most of the effect to the shared head.

Inference and cost

Inference

- run the first k blocks, apply the same head, match by cosine similarity
- k is **selected by the compute budget**, not per input
- the deployed network is a **standard ViT**: no extra layers, no overhead

Training

- both exits come from **one forward pass**
- $+3$-5% training time ($65.5 \rightarrow 68.6$ A100 GPU-h, ViT-S/12)

ViT-S/12, batch 1, RTX 4060 Ti

exit k	GFLOPs	ms
3	4.70	1.52
6	9.26	2.24
9	13.82	2.99
12	18.37	3.75

Experimental setup

- **Data:** WebFace4M; WebFace12M for scaling
- **Backbones:** ViT-S/12, ViT-S/15, ViT-B/24
- **Training:** AdaFace, PartialFC (0.6), AdamW, 40 epochs, batch 1024, $\omega = 0.7$
- **Baseline:** identical backbone and recipe, final exit only ($\omega = 1$)

Benchmarks: LFW, AgeDB-30, CFP-FP, CPLFW, CALFW (HQ (5) average), TinyFace Rank-1, IJB-C TAR@FAR$= 10^{-4}$

Three protocols

- **self:** probe and gallery at exit k
- **gallery-fixed:** exit-k probe, gallery enrolled at full depth
- **one threshold:** τ calibrated at full depth, fixed for every exit

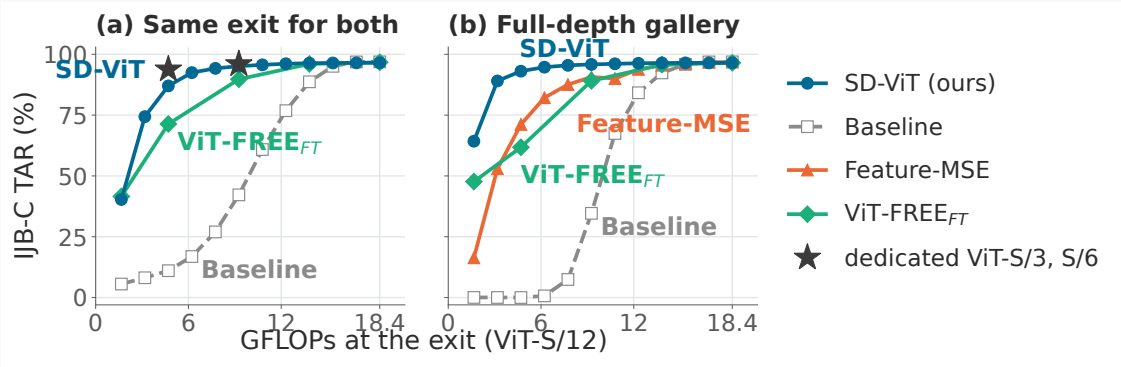
Every depth becomes a usable exit

Self-evaluation, IJB-C TAR@FAR$= 10^{-4}$ at quarter / half / full depth.

Backbone, data	Method	$\frac{1}{4}$	$\frac{1}{2}$	full
ViT-S/12, WF4M	Baseline	11.05	42.22	96.89
	SD-ViT	**87.01**	**95.11**	96.51
ViT-B/24, WF4M	Baseline	21.63	71.97	97.02
	SD-ViT	**91.83**	**96.32**	96.69
ViT-B/24, WF12M	Baseline	12.10	55.93	97.69
	SD-ViT	**95.03**	**97.06**	97.56

Over **96% of full-depth performance at 50% of the compute**; full depth stays **within 0.4 IJB-C points** of the identically trained Baseline.

At every exit, on both protocols



ViT-S/12, IJB-C TAR at FAR 10^{-4}, every exit. **(a)** probe and gallery at the same exit; **(b)** probe at exit k against a gallery enrolled once at full depth. Stars: dedicated ViT-S/3 and ViT-S/6, each trained from scratch for one budget.

Against a gallery enrolled once at full depth

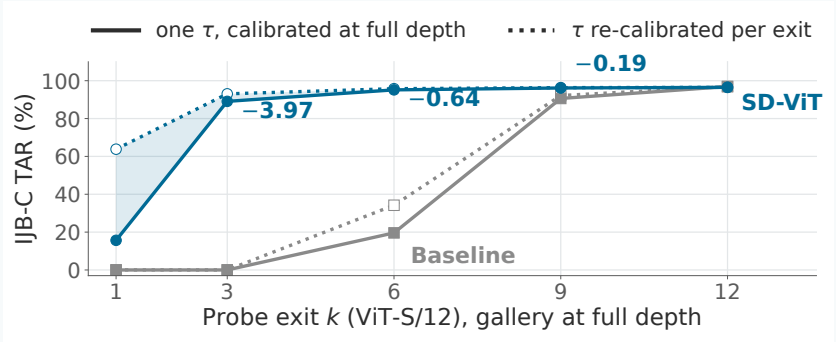
ViT-S/12, WF4M, IJB-C TAR@FAR$= 10^{-4}$, exit-k probe vs. full-depth gallery.

exit k	1	3	6	9	12
Baseline	0.05	0.02	34.63	92.35	96.89
Feature-MSE	16.33	71.16	90.69	95.50	96.82
ViT-FREE$_{FT}$	47.68	61.78	89.09	95.75	96.51
SD-ViT	**64.27**	**93.06**	**95.84**	**96.43**	96.51

TinyFace Rank-1 at quarter depth: **59.31** for SD-ViT against **0.00** for the Baseline.

One verification threshold for every exit

TAR at a target FAR may pick a **different threshold per exit**. We calibrate **one** τ **on full-depth development scores** and apply it unchanged at every exit.



$\tau = 0.382$ (FAR 10^{-4}); labels: TAR lost to the single threshold. Realized FAR stays **below target at every early exit**. Usable **from quarter depth onward**; the Baseline only from exit 9 (90.61).

Against dedicated models and ViT-FREE

Dedicated single-budget ViTs

IJB-C	4.70 GF	9.26 GF
ViT-S/3, ViT-S/6	**93.78**	**95.94**
SD-ViT exit 3, 6	87.01	95.11

Better at their own budget, but N budgets need N **networks and** N **galleries**.

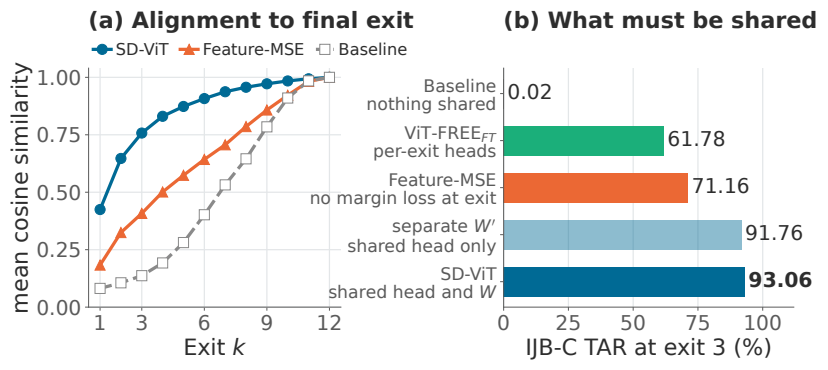
ViT-FREE$_{FT}$, per-exit projections

IJB-C, exit 3	self	gallery-fixed
ViT-FREE$_{FT}$	71.36	61.78
SD-ViT	**87.01**	**93.06**

Our adaptation, trained on WebFace4M on top of the frozen Baseline: each exit gets its **own space**.

The contribution is **coverage**: every budget from one network, against one gallery.

What has to be shared?



- **(a)** SD-ViT's exit 3 is closer to the final embedding (0.76) than an explicit feature-MSE objective gets (0.41).
- **(b)** The exit needs the **margin loss itself**; separate W' costs **about a point**; per-exit heads lose the shared space.

Where, and how many exits, to supervise

Fixed exits peak where they are supervised

- $\{6\}$: 96.07 at exit 6, but **17.9 / 38.1** at exits 2 / 3
- $\{3, 6\}$: 91.89 at exit 3, but **48.7** at exit 2
- sampling: 74.4 / 87.0 at exits 2 / 3

Fixed exits win only if the budgets are known before training and never change.

All exits, deterministically

IJB-C	sampled	all 11
exit 3	87.01	86.54
exit 6	95.11	95.15
A100 GPU-h	**68.6**	88.3

Sampling is a **constant-cost estimator** of all-exit deep supervision: on par, at 1/7.4 of the auxiliary cost.

Ablations

IJB-C TAR@FAR= 10^{-4} at half depth, WebFace4M.

Change	Setting	Baseline	SD-ViT
–	ViT-S/12, AdaFace (default)	42.22	**95.11**
margin loss	ArcFace	57.82	**94.97**
depth	ViT-S/15	70.81	**95.59**
backbone	KP-RPE	53.52	**95.27**

- **Weight ω:** full depth varies by at most 0.53 IJB-C over $\omega \in \{0.5, 0.7, 0.9\}$; early exits collapse only at $\omega = 1$ (backup).

Scope

- **Budget-selected fixed-depth exits:** a per-image exit policy is future work.
- **Constant token width:** one head must decode every block; verified on a plain ViT and on KP-RPE.
- **Within one model:** probe and gallery come from the same network.
- **One seed per configuration:** differences of a few tenths are not significant.

Take-aways

- Early-exit FR needs **gallery consistency**: one embedding space across depth.
- SD-ViT gets it by supervising a **sampled block through the final head and prototypes**.
- One training run gives **a usable exit at every block depth**, one gallery, and **one threshold** from quarter depth onward.

code and models

github.com/gurayozgur/SD-ViT

Thank you

Questions?



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Backup: one threshold, all methods

IJB-C, gallery-fixed, FAR target 10^{-4}; TAR with τ fixed at full depth / re-calibrated per exit.

exit k	1	3	6	9	12
Baseline	0.00 / 0.04	0.00 / 0.01	19.59 / 34.19	90.61 / 92.16	96.91
Feature-MSE	0.00 / 16.69	29.24 / 70.80	79.56 / 90.48	94.36 / 95.47	96.78
SD-ViT	15.66 / 63.81	89.08 / 93.05	95.15 / 95.78	96.15 / 96.34	96.48

Realized FAR at the fixed τ for SD-ViT: 0.01, 0.13, 0.43, 0.75, 1.02×10^{-4} at exits 1, 3, 6, 9, 12. Shallower probes lower the genuine mean far more than the impostor mean, so the fixed threshold drifts conservative.

Backup: primary-loss weight ω

ViT-S/12, WF4M, IJB-C TAR@FAR$= 10^{-4}$.

ω	0.0	0.5	**0.7**	0.9	1.0 (Baseline)
exit 6	95.21	95.52	95.11	93.35	42.22
exit 12	95.25	96.20	96.51	96.73	96.89

A dominant primary term is needed ($\omega = 0$ costs full depth), and early exits collapse only at $\omega = 1$.

Backup: training cost

ViT-S/12, WF4M	A100 GPU-h	over Baseline
Baseline	65.5	-
SD-ViT (one sampled exit)	68.6	+3.1
all 11 exits, deterministic	88.3	+22.8
KD, teacher MSE	84.7	+19.2

ViT-B/24 on WebFace12M: 411.1 $\rightarrow$ 430.2 GPU-h. The deployed network is unchanged, so inference cost is identical to the Baseline.